Solving Standard Minimization Problems

- Reconstruct the **standard minimization problem** into **a standard maximization problem** by converting it to a **Dual Problem**.
- The original minimum problem is called the **Primal Problem.**
- The corresponding maximum problem is called the **Dual Problem.**
- Then we solve the dual problem using the **Simplex Method.**
- The **maximum of the dual problem** is also the **minimum of the primal problem.**
- The solution to the **maximum problem** is found in the **last column**, but
- The solution to the **minimum problem** is found in the in the **last row**.

Formation of the Dual Problem:

Step 1. Use the **coefficients** and **constants** in the problem constraints and the objective function to form a **matrix A** with the coefficients of the objective function in the last row.

Step 2. Interchange the **rows** and **columns** of **matrix A** to form the **matrix A^T** *(transpose of A)*.

Step 3. Use the **rows of A^T to form a maximization problem** with problem constraints.

Consider the following minimization problem:

Minimize $C = 16X_1 + 45X_2$

Subject to: $2X_1 + 5X_2 \geq 50$

$X_1 + 3X_2 \geq 27$

$X_1 , X_2 \geq 0$

Matrix form of the problem:

$$A = \left(\begin{array}{cc|c} 2 & 5 & 50 \\ 1 & 3 & 27 \\ \hline 16 & 45 & 1 \end{array}\right)$$

- The transpose of the matrix can be obtained by interchanging rows with its corresponding columns

$$A = \left(\begin{array}{cc|c} 2 & 5 & 50 \\ 1 & 3 & 27 \\ \hline 16 & 45 & 1 \end{array}\right) \quad \Rightarrow \quad A^T = \left(\begin{array}{cc|c} 2 & 1 & 16 \\ 5 & 3 & 45 \\ \hline 50 & 27 & 1 \end{array}\right)$$

Primal Problem

Minimize: $C = 16X_1 + 45X_2$

Subject to: $2X_1 + 5X_2 \geq 50$

$X_1 + 3X_2 \geq 27$

$X_1 , X_2 \geq 0$

Dual Problem

Maximize: $P = 50Y_1 + 27Y_2$

Subject to: $2Y_1 + Y_2 \leq 16$

$5Y_1 + 3Y_2 \leq 45$

$Y_1 , Y_2 \geq 0$

Consider the Dual Problem

Maximize : $P = 50Y_1 + 27Y_2 \rightarrow P - 50Y_1 - 27Y_2 = 0$

Subject to: $2Y_1 + Y_2 + X_1 = 16$

$5Y_1 + 3Y_2 + X_2 = 45$

$Y_1, Y_2, X_1, X_2 \geq 0$

X_1 & X_2 are treated slack variables

Initial Tableau:

Basic	P	Y_1	Y_2	X_1	X_2	RHS
X_1	0	2	1	1	0	16
X_2	0	5	3	0	1	45
P	1	-50	-27	0	0	0

Final Simplex Tableau:

Basic	P	Y_1	Y_2	X_1	X_2	RHS
Y_1	0	1	0	3	-1	3
Y_2	0	0	1	-5	2	10
P	1	0	0	15	4	420

Optimal Solution of the Primal Problem

- The optimal solution to **minimization problem** always can be found in the **bottom row** of the final simplex tableau for the dual problem.

Ergo,

$X_1 = 15$
$X_2 = 4$
Minimum Cost: 420

Example : Diet Plan

Assume your diet is based on the two “basic foods” : food A and food B. Food A costs $3 while Food B costs $1.50. Each day, you must have at least 1500 calories, 10 ounce of sugar, and 0.5 ounce of caffeine. If the number of calories for Food A and Food B are 500 and 250, respectively, the sugar contents (in ounce) for Food A and Food B are 0 and 2, while the caffeine contents are 0 and 0.2, respectively, then what is the minimum cost incurred to meet the daily nutrient requirement?

Continuation…

Assume your diet is based on the two "basic foods" : food A and food B. Food A costs $3 while Food B costs $1.50. Each day, you must have at least 1500 calories, 10 ounce of sugar, and 0.5 ounce of caffeine. If the number of calories for Food A and Food B are 500 and 250, respectively, the sugar contents (in ounce) for Food A and Food B are 0 and 2, while the caffeine contents are 0 and 0.2, respectively, then what is the minimum cost incurred to meet the daily nutrient requirement?

	Food A	Food B	Limitation (At least)
Calories	500	250	1500
Sugar	0	2	10
Caffeine	0	0.2	0.5
Cost	$3	$1.5	

Continuation…

Let : X_1 = number of pieces of Food A

X_2= number of pieces of Food B

C = cost

Objective function: Min C = $3X_1 + 1.5X_2$

Calories content constraint

$$500X_1 + 250X_2 \geq 1500$$

Sugar content constraint

$$0X_1 + 2X_2 \geq 10$$

Caffeine content constraint

$$0X_1 + 0.2X_2 \geq 0.5$$

Nonnegative constraints

$$X_1, X_2 \geq 0$$

Continuation…

Min C = $3X_1 + 1.5X_2$

Subject to:

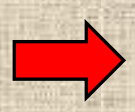
$$500X_1 + 250X_2 \geq 1500$$
$$0X_1 + 2X_2 \geq 10$$
$$0X_1 + 0.2X_2 \geq 0.5$$
$$X_1, X_2 \geq 0$$

Matrix form of the problem:

$$A = \left(\begin{array}{cc|c} 500 & 250 & 1500 \\ 0 & 2 & 10 \\ 0 & 0.2 & 0.5 \\ \hline 3 & 1.5 & 1 \end{array}\right)$$

Continuation…

$$A = \left(\begin{array}{ccc|c} 500 & 250 & & 1500 \\ 0 & 2 & & 10 \\ 0 & 0.2 & & 0.5 \\ \hline 3 & 1.5 & & 1 \end{array}\right)$$



$$A^T = \left(\begin{array}{ccc|c} 500 & 0 & 0 & 3 \\ 250 & 2 & 0.2 & 1.5 \\ \hline 1500 & 10 & 0.5 & 1 \end{array}\right)$$

Primal Problem

Min C = $3X_1 + 1.5X_2$

Subject to:

$$\begin{aligned} 500X_1 + 250X_2 &\geq 1500 \\ 0X_1 + 2X_2 &\geq 10 \\ X_1 + 0.2X_2 &\geq 0.5 \\ X_1, X_2 &\geq 0 \end{aligned}$$

Dual Problem

Max P = $1500Y_1 + 10Y_2 + 0.5Y_3$

Subject to:

$$\begin{aligned} 500Y_1 &\leq 3 \\ 250Y_1 + 2Y_2 + 0.2Y_3 &\leq 1.5 \\ Y_1, Y_2, Y_3 &\geq 0 \end{aligned}$$

Continuation…

Initial Simplex Tableau: (Dual Problem)

Max P = $1500Y_1 + 10Y_2 + 0.5Y_3$

Subject to:

$$500Y_1 + \boldsymbol{X_1} \leq 3$$

$$250Y_1 + 2Y_2 + 0.2Y_3 + \boldsymbol{X_2} \leq 1.5$$

$$X_1, X_2, Y_1, Y_2, Y_3 \geq 0$$

Basic	P	Y_1	Y_2	Y_3	X_1	X_2	RHS
X_1	0	500	0	0	1	0	3
X_2	0	250	2	0.2	0	1	1.5
P	1	–1500	–10	–0.5	0	0	0

Final Simplex Tableau:

Basic	P	Y_1	Y_2	Y_3	X_1	X_2	RHS
Y_1	0	1	0	0	0.002	0	0.006
Y_2	0	0	1	0.1	–0.25	0.5	0
P	1	0	0	0.5	0.5	5	9

Examining the bottom row of the final simplex tableau, we can find the optimal solution to the minimization problem.

$\mathbf{X_1}$ **= 0.5** *(half piece of Food A)*

$\mathbf{X_2}$ **= 5** *(5 pieces of Food B)*

Minimum Cost: $9